Lecture 5

The Beta–Bernoulli Conjugate Model

Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Ber}(\theta), \qquad \theta \sim U(0,1).$$

Then

$$p(\theta \mid x_1, \dots, x_n) \sim \text{Beta}\left(\sum X_i + 1,\ n - \sum X_i + 1\right).$$

$$\boxed{p(\theta \mid x_1 \dots x_n) \ \propto\ \theta^{\sum X_i} (1-\theta)^{n - \sum X_i}}$$

$$\text{Beta}(a,b) \ \propto\ \theta^{a-1}(1-\theta)^{b-1}$$

Conjugate prior. If the prior distribution is of the same family as the posterior distribution, then we call such a prior a **conjugate prior**.

$$p(\theta \mid X_1 \dots X_n) \ \propto\ p(\theta)\, p(X_1 \dots X_n \mid \theta)$$

$$p(\theta \mid X_1 \dots X_n) \ \propto\ \theta^{\alpha + \sum X_i - 1}(1-\theta)^{\beta + (n - \sum X_i) - 1}$$

$$\therefore \quad \text{Beta}\left(\sum X_i + \alpha,\ n - \sum X_i + \beta\right)$$

$$p(\tilde{y} = 1 \mid X_1 \dots X_n) = E(\theta \mid X_1 \dots X_n) = \frac{\alpha + \sum X_i}{n + \alpha + \beta}$$

$$\text{As } n \to \infty, \quad \approx \frac{\sum X_i}{n}$$

General setup. $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Ber}(\theta)$. If $p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$ [Prior], then $\theta \sim \text{Beta}(\alpha, \beta)$.

$$\boxed{\frac{\sum X_i}{n}} \qquad \text{Var}(\theta \mid X_1 \dots X_n) = \frac{(\alpha + \sum X_i)(\beta + n - \sum X_i)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}$$

$$\text{As } n \to \infty, \quad \text{Var}(\theta \mid X_1 \dots X_n) \ \approx\ \frac{(\sum X_i)(n - \sum X_i)}{n^2(n)} \ \longrightarrow\ 0$$

Normal Distribution with Known Variance

$$p(y \mid \theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$$p(\theta \mid y) \ \propto\ \underline{\underline{p(\theta)}}\, \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$$\exp\big(A\theta^2 + B\theta + C\big)\, \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

$p(\theta) \sim N(\mu_0, \tau_0^2)$, & then,

$$p(\theta \mid y) \ \propto\ \exp\left(-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right) \exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)$$

Show:

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where}$$
$$\mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{1}{\sigma^2} y}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \qquad \text{and} \qquad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

Now let $y_1, \ldots, y_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$.

$$\bar{y} \sim N\left(\theta, \frac{\sigma^2}{n}\right)$$
$$p(\theta \mid \bar{y}) \ \propto \ p(\theta)\, p(\bar{y} \mid \theta)$$
$$\therefore \ \theta \mid y_1, \ldots, y_n \ (\text{equivalently } \theta \mid \bar{y})$$
$$\therefore \ \theta \mid \bar{y} \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \qquad \& \qquad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

$$p(y \mid \theta) \ \propto \ \exp\left\{-\frac{\sum (y_i - \theta)^2}{2\sigma^2}\right\} \qquad \exp\left\{-\frac{\left(\frac{\sum y_i}{n} - \theta\right)^2}{2\sigma^2/n}\right\}$$

$p(\theta)$ is $N(\underline{\mu_0}, \tau_0^2)$ $\qquad p(\theta \mid y_1, \ldots, y_n) \quad N(\mu_1, \tau_1^2)$

$\bar{y}$ is an estimator of θ.

Take $n = 1$:

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \qquad \boxed{\bar{y} = \bar{y}} \qquad \text{Precision} = \frac{1}{\text{Variance}}$$

- $\boxed{\mu_1 \equiv \mu_0}$ if $y = \mu_0$ or $\tau_0^2 = 0$ $\quad (\& \ \sigma^2 \neq 0)$
- $\mu_1 = y$ if $y = \mu_0$ or $\sigma^2 = 0, \ \tau^2 \neq 0$
- μ_1 is the weighted average of μ_0 & y.
- Let's take $\mu_0 = 0$, $\quad \mu_1 = \dfrac{y\,\tau_0^2}{\sigma^2 + \tau_0^2}$

$$\mu_1 = \frac{y\,\tau_0^2/n}{\sigma^2/n + \tau_0^2}$$

$$\tau_0^2(n) = \frac{\sigma^2}{n}$$

$$p(\theta \mid y) \ \propto \ p(\theta) \prod_{i=1}^{n} p(y_i \mid \theta)$$

Predictive Distribution

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)\, p(\theta \mid y)\, d\theta$$

$$\propto \int \exp\left(-\frac{1}{2\sigma^2}(\tilde{y} - \theta)^2\right) \exp\left(-\frac{1}{2\tau_1^2}(\theta - \mu_1)^2\right) d\theta$$

pdf of bivariate normal $(\tilde{y}, \theta)$.

Q: $p(y) = ?$ (Marginal of y)

$$\theta \sim N(\mu_0, \tau_0^2), \qquad y \sim N(\theta, \sigma^2) \quad \text{then what is } p(y)?$$

Lecture 6

Recap and the Marginal Distribution of y

$$(\cdot\ \ldots X_n) \qquad \theta \mid x_1 \ldots x_n \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

$$p(y, \underline{\theta}) \propto \exp\left\{ - \left(\begin{pmatrix} \theta \\ y \end{pmatrix}^T - \mu \right)^T \Sigma^{-1} \left(\begin{pmatrix} \theta \\ y \end{pmatrix} - \mu \right) \right\}$$

$$\theta \sim N(\mu_0, \tau_0^2), \qquad y \mid \theta \sim N(\theta, \sigma^2)$$

$$\text{then} \quad p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta = \int \underbrace{\frac{\exp\left(-\frac{1}{2\sigma^2}(y-\theta)^2\right)}{\sigma\sqrt{2\pi}} \cdot \frac{\exp\left(-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2\right)}{\tau_0\sqrt{2\pi}}} \, d\theta$$

$$y \mid \theta \sim N(\theta, \sigma^2)$$
$$y \mid \theta = \theta + N(0, \sigma^2)$$
$$y = \theta + \varepsilon \qquad = \int p(y, \theta)\, d\theta$$

Marginal distribution of $\theta \sim N(\mu_0, \tau_0^2)$, y is Normal distribution

$$\boxed{y \sim N(\mu_0,\ \tau_0^2 + \sigma^2)}$$

$y = \theta + \varepsilon$ where $\theta \sim N(\mu_0, \tau_0^2)$.

$$\begin{pmatrix} \theta \\ \varepsilon \end{pmatrix} \text{ is a bivariate normal r.v.} \qquad \varepsilon \sim N(0, \sigma^2) \ \ \& \ \ \theta, \varepsilon \text{ are independent.}$$

The Joint (Multivariate Normal) Formulation

Let $y_1, \ldots, y_n \mid \theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$ & $\theta \sim N(\mu_0, \tau_0^2)$. What is $p(y_1, \ldots, y_n)$?

$$y_1 = \theta + \varepsilon_1, \quad \ldots, \quad y_n = \theta + \varepsilon_n$$

$$\left.\varepsilon_1, \ldots, \varepsilon_n \text{ are iid } N(0, \sigma^2)\right\} \text{ independent of } \theta \sim N(\mu_0, \tau_0^2)$$

$$\begin{pmatrix} \theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix} \sim MVN(\mathcal{M},\ \Sigma), \qquad \mathcal{M} = \begin{pmatrix} \mu_0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \tau_0^2 & 0 & & & \\ 0 & \sigma^2 & 0 & & \\ & 0 & \sigma^2 & 0 & \\ & & 0 & \sigma^2 & 0 \\ & & & & \ddots \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = A \begin{pmatrix} \theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 & 0 & 0 & \cdots \\ 1 & 0 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 1 & \cdots \\ \vdots & & & & \ddots \end{pmatrix}, \qquad y = Ax$$

$$y = AX \implies y \sim MN\big(A\mathcal{M},\ A\Sigma A^T\big)$$

$$\text{Cov}(\theta + \varepsilon_1,\ \theta + \varepsilon_2) = \tau^2$$

$$\boxed{\theta \mid y_1, \ldots, y_n \sim N(\mu_1, \tau_1^2)}$$

$$y \mid \theta \sim N(\theta, \sigma^2) \qquad \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim MN\left(\begin{pmatrix} \mu_0 \\ \vdots \\ \mu_0 \end{pmatrix},\ \begin{pmatrix} \tau^2 + \sigma^2 & \tau^2 & \cdots \\ \tau^2 & \tau^2 + \sigma^2 & \\ \vdots & & \ddots \end{pmatrix} \right)$$

$$y \mid \theta = \underline{\underline{Z}} + \underline{\theta}, \qquad Z \sim N(0, \sigma^2)$$

Prediction for Unobserved Data

$$\theta \sim N(\mu_0, \tau_0^2)$$

$$y_1, \ldots, y_M \mid \theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$$

Given $y_1, \ldots, y_n$, with y_M unobserved:

$$p(y_M \mid y_1, \ldots, y_n) = \int \underline{p(y_M \mid \theta)}\ \underline{\underline{p(\theta \mid y_1, \ldots, y_n)}}\, d\theta \qquad y_M \mid y_1, \ldots, y_n \sim N\big(\mu_1,\ \sigma^2 + \tau_1^2\big)$$

$$\int p(y \mid \theta)\, \underline{\underline{p(\theta)}}\, d\theta \qquad y_{n+1}, \ldots, y_M \mid \underbrace{y_1, \ldots, y_n} \longrightarrow \sum_{k=1}^{M-n} y_{n+k}$$

$$\tau_1^2 = \frac{1}{\dfrac{1}{\tau_0^2} + \dfrac{n}{\sigma^2}} \longrightarrow 0 \qquad \Big(\text{so } \ p(y_1, \ldots, y_n) \longrightarrow \delta_{\theta_0} \ \text{ as } n \to \infty\Big)$$

$$y_{n+1}, \ldots y_M \mid y_1, \ldots y_n \rightarrow \Sigma y_i$$

$$\sum_{k=1}^{m-n} y_{n+k}$$

$$\begin{pmatrix} \tau_1^2 + \sigma_1^2 & \tau_1^2 & - & - \\ \tau_1^2 & \tau_1^2 + \sigma_1^2 & & \\ \vdots & & \ddots & \\ & & & \tau_1^2 + \sigma_1^2 \end{pmatrix}$$

The Beta–Bernoulli Model Revisited (General α, β)

$$X_1, \ldots, X_n \mid \theta \overset{\text{iid}}{\sim} \text{Ber}(\theta), \qquad \theta \sim \text{Beta}(\alpha, \beta)$$
$$P(X_1 = 1) = \frac{\alpha}{\alpha + \beta}$$

$$P(X_1, \ldots, X_n) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{\sum X_i + \alpha - 1} (1-\theta)^{n - \sum X_i + \beta - 1} \, d\theta = \frac{B(\alpha + \sum X_i,\ \beta + n - \sum X_i)}{B(\alpha, \beta)}$$

A Note on Independence

$$P(X_1 = 1,\, X_2 = 1) \ \neq \ P(X_1 = 1)\, P(X_2 = 1)$$

$$\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{array}$$

Sampling algorithms (mentioned in passing):

- Metropolis–Hastings algorithm
- MCMC (Markov Chain Monte Carlo)
- HMC (Hamiltonian Monte Carlo)

Exponential Family of Distributions

$$p(y_i \mid \theta) = f(y_i)\, g(\theta)\, \exp\left\{\phi(\theta)^T u(y_i)\right\}. \qquad \text{e.g. Poisson, Normal, Binomial, Exponential.}$$

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} f(y_i)\, (g(\theta))^n \, \exp\left\{\phi(\theta)^T \sum_{i=1}^{n} u(y_i)\right\}.$$

$$p(\theta \mid y) \propto p(\theta)[(g(\theta))^n \exp\left\{\phi(\theta)^T \sum_i u(y_i)\right\}].$$

$$p(\theta) \propto (g(\theta))^\eta \exp\left\{\phi(\theta)^T \nu\right\}.$$

$$\text{Then,} \qquad p(\theta \mid y) \propto (g(\theta))^{\eta+n} \exp\left\{\phi(\theta)^T \left(\sum_i u(y_i) + \nu\right)\right\}.$$

Show that the Binomial is an exponential family of distribution. Hence put an appropriate conjugate prior over it.

$$\text{Beta}(\alpha, \beta) \qquad \boxed{\frac{\alpha}{\alpha+\beta}}$$

$$\left[\text{Natural parameter is } \log(\theta/(1-\theta))\right]$$